

Importing all needed libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Loading Dataset

```
data=pd.read_csv(r"datasets.csv",encoding_errors='ignore')
```

Initial Exploration

```
data.head()
```

	id	name
host_id \		
0	1.312228e+06	Rental unit in Brooklyn · ★5.0 · 1 bedroom 7130382
1	4.527754e+07	Rental unit in New York · ★4.67 · 2 bedrooms · ... 51501835
2	9.710000e+17	Rental unit in New York · ★4.17 · 1 bedroom · ... 528871354
3	3.857863e+06	Rental unit in New York · ★4.64 · 1 bedroom · ... 19902271
4	4.089661e+07	Condo in New York · ★4.91 · Studio · 1 bed · 1... 61391963

	host_name	neighbourhood_group	neighbourhood
latitude \			
0	Walter	Brooklyn	Clinton Hill 40.683710
1	Jeniffer	Manhattan	Hell's Kitchen 40.766610
2	Joshua	Manhattan	Chelsea 40.750764
3	John And Catherine	Manhattan	Washington Heights 40.835600
4	Stay With Vibe	Manhattan	Murray Hill 40.751120

	longitude	room_type	price	...	last_review
reviews_per_month \					
0	-73.964610	Private room	55.0	...	20-12-2015 0.03
1	-73.988100	Entire home/apt	144.0	...	01-05-2023 0.24
2	-73.994605	Entire home/apt	187.0	...	18-12-2023 1.67

```

3 -73.942500 Private room 120.0 ... 17-09-2023
1.38
4 -73.978600 Entire home/apt 85.0 ... 03-12-2023
0.24

```

```

    calculated_host_listings_count availability_365
number_of_reviews_ltm \
0                                1.0            0.0
0.0
1                                139.0           364.0
2.0
2                                1.0            343.0
6.0
3                                2.0            363.0
12.0
4                                133.0           335.0
3.0

```

```

    license rating bedrooms beds baths
0 No License      5         1    1 Not specified
1 No License    4.67         2    1             1
2      Exempt    4.17         1    2             1
3 No License    4.64         1    1             1
4 No License    4.91   Studio    1             1

```

[5 rows x 22 columns]

```
data.tail()
```

```

            id name
\
20765  2.473690e+07 Rental unit in New York · ★4.75 · 1 bedroom · ...
20766  2.835711e+06 Rental unit in New York · ★4.46 · 1 bedroom · ...
20767  5.182527e+07 Rental unit in New York · ★4.93 · 1 bedroom · ...
20768  7.830000e+17 Rental unit in New York · ★5.0 · 1 bedroom · 1...
20769  5.660000e+17 Rental unit in Queens · ★4.89 · 1 bedroom · 1 ...

```

```

    host_id host_name neighbourhood_group neighbourhood
latitude \
20765  186680487   Henry D           Manhattan  Lower East Side
40.711380
20766   3237504     Aspen           Manhattan  Greenwich Village
40.730580
20767  304317395      Jeff           Manhattan  Hell's Kitchen
40.757350
20768  163083101   Marissa           Manhattan  Chinatown

```

40.713750				
20769	93827372	Glenroy	Queens	Rosedale
40.658874				

	longitude	room_type	price	...	last_review
reviews_per_month 20765 1.81	-73.991560	Private room	45.0	...	29-09-2023
20766 0.48	-74.000700	Entire home/apt	105.0	...	01-07-2023
20767 2.09	-73.993430	Entire home/apt	299.0	...	08-12-2023
20768 0.91	-73.991470	Entire home/apt	115.0	...	17-09-2023
20769 4.50	-73.728651	Private room	102.0	...	10-12-2023

calculated_host_listings_count	availability_365
20765	157.0
12.0	
20766	0.0
1.0	
20767	0.0
27.0	
20768	363.0
7.0	
20769	0.0
62.0	

		license	rating	bedrooms	beds	baths
20765	No License		4.75	1	1	1
20766	No License		4.46	1	2	1
20767	No License		4.93	1	1	1
20768	No License		5	1	1	1
20769	OSE-STRREG-0000513		4.89	1	1	1

```
[5 rows x 22 columns]
```

```
data.shape
```

(20770, 22)

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 20770 entries, 0 to 20769
```

Data columns (total 22 columns):

#	Column	Non-Null Count	Dtype
0	id	20770 non-null	float64

1	name	20770	non-null	object
2	host_id	20770	non-null	int64
3	host_name	20770	non-null	object
4	neighbourhood_group	20770	non-null	object
5	neighbourhood	20763	non-null	object
6	latitude	20763	non-null	float64
7	longitude	20763	non-null	float64
8	room_type	20763	non-null	object
9	price	20736	non-null	float64
10	minimum_nights	20763	non-null	float64
11	number_of_reviews	20763	non-null	float64
12	last_review	20763	non-null	object
13	reviews_per_month	20763	non-null	float64
14	calculated_host_listings_count	20763	non-null	float64
15	availability_365	20763	non-null	float64
16	number_of_reviews_ltm	20763	non-null	float64
17	license	20770	non-null	object
18	rating	20770	non-null	object
19	bedrooms	20770	non-null	object
20	beds	20770	non-null	int64
21	baths	20770	non-null	object

dtypes: float64(10), int64(2), object(10)

memory usage: 3.5+ MB

Statistical Summary

data.describe()

	id	host_id	latitude	longitude
price \				
count	2.077000e+04	2.077000e+04	20763.000000	20763.000000
20736.000000				
mean	3.033858e+17	1.749049e+08	40.726821	-73.939179
187.714940				
std	3.901221e+17	1.725657e+08	0.060293	0.061403
1023.245124				
min	2.595000e+03	1.678000e+03	40.500314	-74.249840
10.000000				
25%	2.707260e+07	2.041184e+07	40.684159	-73.980755
80.000000				
50%	4.992852e+07	1.086990e+08	40.722890	-73.949597
125.000000				
75%	7.220000e+17	3.143997e+08	40.763106	-73.917475
199.000000				
max	1.050000e+18	5.504035e+08	40.911147	-73.713650
100000.000000				

	minimum_nights	number_of_reviews	reviews_per_month \
count	20763.000000	20763.000000	20763.000000
mean	28.558493	42.610605	1.257589
std	33.532697	73.523401	1.904472

min	1.000000	1.000000	0.010000
25%	30.000000	4.000000	0.210000
50%	30.000000	14.000000	0.650000
75%	30.000000	49.000000	1.800000
max	1250.000000	1865.000000	75.490000

	calculated_host_listings_count	availability_365	\
count	20763.000000	20763.000000	
mean	18.866686	206.067957	
std	70.921443	135.077259	
min	1.000000	0.000000	
25%	1.000000	87.000000	
50%	2.000000	215.000000	
75%	5.000000	353.000000	
max	713.000000	365.000000	

	number_of_reviews_ltm	beds
count	20763.000000	20770.000000
mean	10.848962	1.723592
std	21.354876	1.211993
min	0.000000	1.000000
25%	1.000000	1.000000
50%	3.000000	1.000000
75%	15.000000	2.000000
max	1075.000000	42.000000

Data Cleaning

```
data.isnull().sum()
```

id	0
name	0
host_id	0
host_name	0
neighbourhood_group	0
neighbourhood	7
latitude	7
longitude	7
room_type	7
price	34
minimum_nights	7
number_of_reviews	7
last_review	7
reviews_per_month	7
calculated_host_listings_count	7
availability_365	7
number_of_reviews_ltm	7
license	0
rating	0
bedrooms	0

```
beds 0
baths 0
dtype: int64
```

```
data.dropna(inplace=True)
```

```
data.isnull().sum()
```

```
id 0
name 0
host_id 0
host_name 0
neighbourhood_group 0
neighbourhood 0
latitude 0
longitude 0
room_type 0
price 0
minimum_nights 0
number_of_reviews 0
last_review 0
reviews_per_month 0
calculated_host_listings_count 0
availability_365 0
number_of_reviews_ltm 0
license 0
rating 0
bedrooms 0
beds 0
baths 0
dtype: int64
```

```
data.shape
```

```
(20736, 22)
```

```
data.duplicated().sum()
```

```
np.int64(12)
```

```
data[data.duplicated()]
```

	id	name
\		
6	4.527754e+07	Rental unit in New York · ★4.67 · 2 bedrooms · ...
7	9.710000e+17	Rental unit in New York · ★4.17 · 1 bedroom · ...
8	3.857863e+06	Rental unit in New York · ★4.64 · 1 bedroom · ...
9	4.089661e+07	Condo in New York · ★4.91 · Studio · 1 bed · 1...

10	4.958498e+07	Rental unit in New York · ★5.0 · 1 bedroom · 1...
20736	7.990000e+17	Rental unit in New York · 2 bedrooms · 2 beds ...
20737	5.930000e+17	Rental unit in New York · ★4.79 · 2 bedrooms · ...
20738	9.230000e+17	Loft in New York · ★4.33 · 1 bedroom · 2 beds ...
20739	1.336161e+07	Rental unit in New York · ★4.89 · 2 bedrooms · ...
20740	5.119566e+07	Rental unit in New York · Studio · 1 bed · 1 bath
20741	2.523473e+07	Rental unit in New York · ★4.41 · 1 bedroom · ...
20742	3.339399e+06	Rental unit in New York · ★4.73 · 1 bedroom · ...

	host_id	host_name	neighbourhood_group
neighbourhood \			
6 Kitchen	51501835	Jeniffer	Manhattan Hell's
7 Chelsea	528871354	Joshua	Manhattan
8 Heights	19902271	John And Catherine	Manhattan Washington
9 Murray Hill	61391963	Stay With Vibe	Manhattan
10 Kitchen	51501835	Jeniffer	Manhattan Hell's
20736 East Side	224733902	CozySuites Copake	Manhattan Upper
20737 Village	23219783	Rob	Manhattan West
20738 Village	520265731	Rodrigo	Manhattan Greenwich
20739 Harlem	8961407	Jamie	Manhattan
20740 Chinatown	51501835	Jeniffer	Manhattan
20741 East Side	1497427	Mara	Manhattan Upper
20742 Village	2119276	Urban Furnished	Manhattan West

	latitude	longitude	room_type	price	...	last_review
\						
6	40.766610	-73.988100	Entire home/apt	144.0	...	01-05-2023
7	40.750764	-73.994605	Entire home/apt	187.0	...	18-12-2023

8	40.835600	-73.942500	Private room	120.0	...	17-09-2023
9	40.751120	-73.978600	Entire home/apt	85.0	...	03-12-2023
10	40.759950	-73.992960	Entire home/apt	115.0	...	29-07-2023
20736	40.768970	-73.957592	Entire home/apt	153.0	...	15-09-2023
20737	40.730220	-74.002910	Entire home/apt	175.0	...	22-11-2023
20738	40.728390	-73.999540	Entire home/apt	156.0	...	02-01-2024
20739	40.805700	-73.946250	Entire home/apt	397.0	...	08-09-2023
20740	40.718360	-73.995850	Entire home/apt	100.0	...	25-05-2023
20741	40.774030	-73.950580	Entire home/apt	120.0	...	31-03-2023
20742	40.732030	-74.006760	Entire home/apt	143.0	...	09-05-2023
	reviews_per_month	calculated_host_listings_count				
availability_365 \						
6	0.24		139.0			
364.0						
7	1.67		1.0			
343.0						
8	1.38		2.0			
363.0						
9	0.24		133.0			
335.0						
10	0.16		139.0			
276.0						
20736	0.41		8.0			
308.0						
20737	2.03		4.0			
129.0						
20738	2.60		1.0			
356.0						
20739	1.08		3.0			
274.0						
20740	0.08		139.0			
306.0						
20741	0.26		2.0			
364.0						
20742	0.20		54.0			
285.0						
	number_of_reviews_ltm	license	rating	bedrooms	beds	
baths						

6	2.0	No License	4.67	2	1
1					
7	6.0	Exempt	4.17	1	2
1					
8	12.0	No License	4.64	1	1
1					
9	3.0	No License	4.91	Studio	1
1					
10	2.0	No License	5	1	1
1					
20736	2.0	No License	No rating	2	2
2					
20737	25.0	No License	4.79	2	2
1					
20738	9.0	Exempt	4.33	1	2
1					
20739	3.0	No License	4.89	2	2
1					
20740	1.0	No License	No rating	Studio	1
1					
20741	1.0	No License	4.41	1	2
1					
20742	2.0	No License	4.73	1	1
1					

[12 rows x 22 columns]

Deleting all duplicate data

data.drop_duplicates(inplace=True)

data.duplicated().sum()

np.int64(0)

Type Casting: Changing data types

data.dtypes

data['id']=data['id'].astype(object)

data['host_id']=data['host_id'].astype(object)

data['last_review'] = pd.to_datetime(

data['last_review'], # column to convert

format='%Y-%m-%d', # specify format if known, else Pandas

will infer

errors='coerce' # invalid parsing will be set as NaT

)

data.dtypes

id	object
name	object
host_id	object
host_name	object

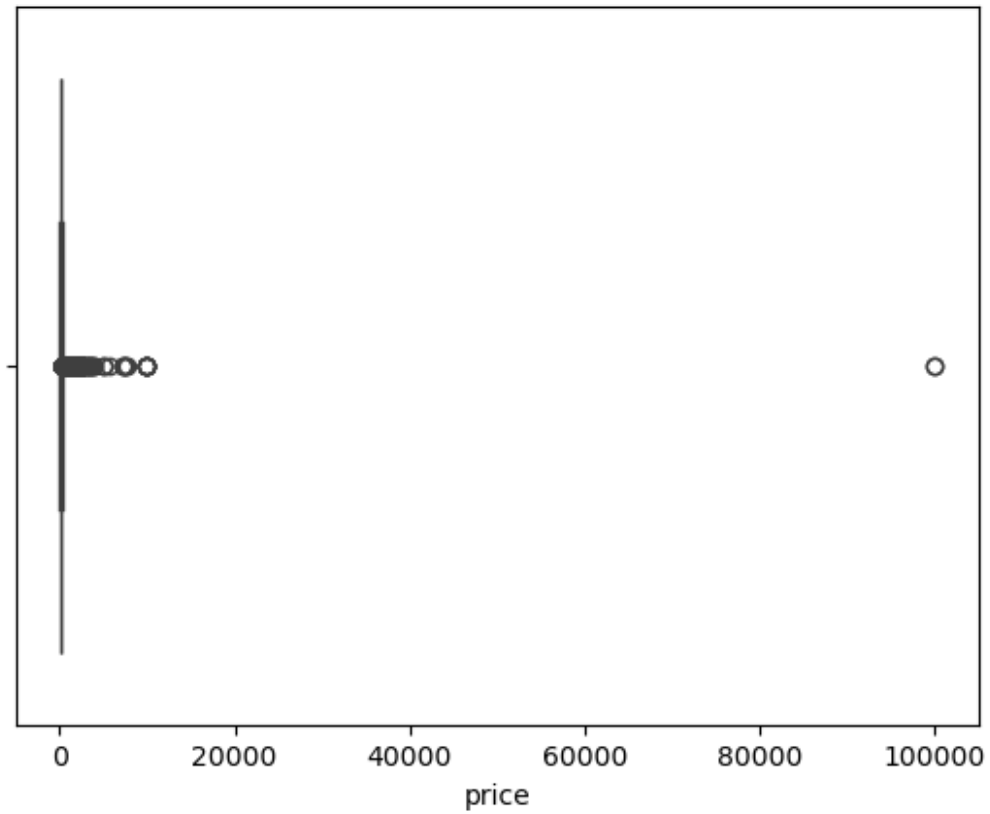
neighbourhood_group	object
neighbourhood	object
latitude	float64
longitude	float64
room_type	object
price	float64
minimum_nights	float64
number_of_reviews	float64
last_review	datetime64[ns]
reviews_per_month	float64
calculated_host_listings_count	float64
availability_365	float64
number_of_reviews_ltm	float64
license	object
rating	object
bedrooms	object
beds	int64
baths	object
dtype:	object

Data Analysis : EDA

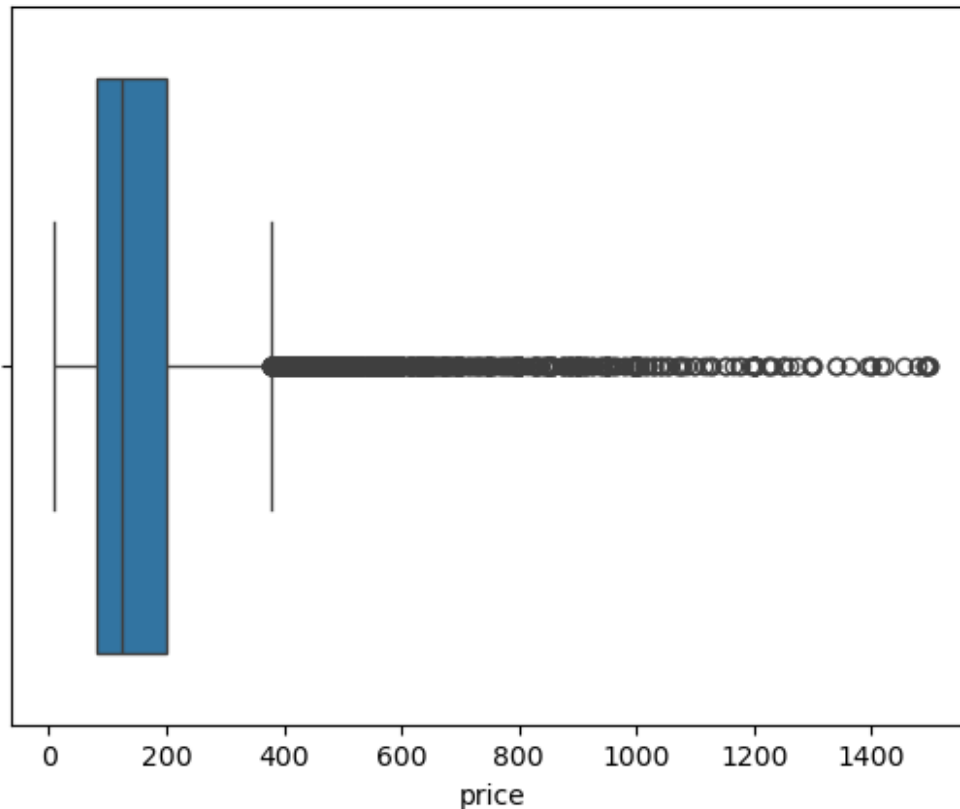
Univariate Analysis

```
# Identifying Outliers in price
sns.boxplot(data=data,x='price')
# px.box(data,x='price')

<Axes: xlabel='price'>
```



```
df=data[data['price']<1500]
# px.box(df,x='price')
sns.boxplot(data=df,x='price')
<Axes: xlabel='price'>
```



```
df.columns
```

```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',
      'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
      'minimum_nights', 'number_of_reviews', 'last_review',
      'reviews_per_month', 'calculated_host_listings_count',
      'availability_365', 'number_of_reviews_ltm', 'license',
      'rating',
      'bedrooms', 'beds', 'baths'],
      dtype='object')
```

```
import math
```

```
columns = [
    'price',
    'beds',
    'availability_365',
    # 'reviews_per_month',
    # 'calculated_host_listings_count',
    # 'number_of_reviews_ltm',
    # 'number_of_reviews'
]
```

```
n_cols = 3 # plots per row
n_rows = math.ceil(len(columns) / n_cols)
```

```

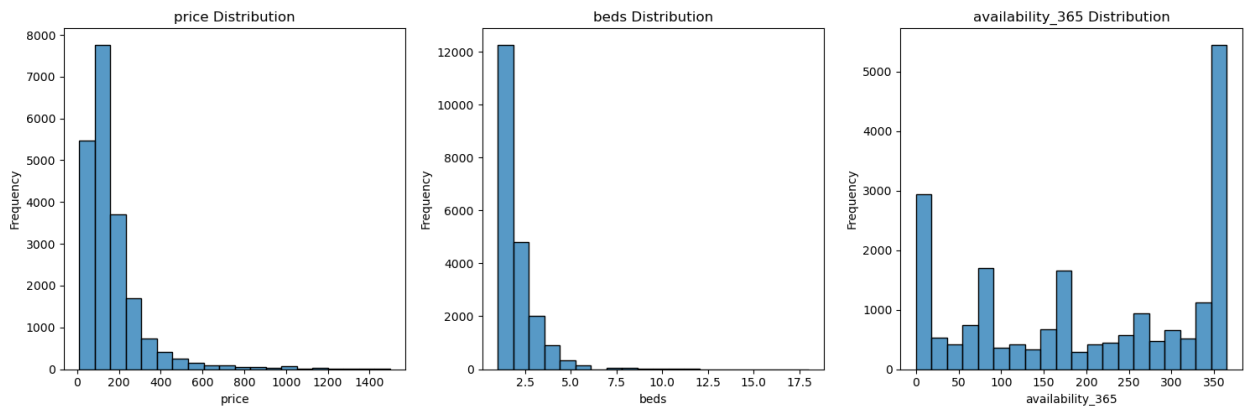
fig, axes = plt.subplots(n_rows, n_cols, figsize=(15, 5 * n_rows))
axes = axes.flatten() # convert to 1D array

for i, col in enumerate(columns):
    sns.histplot(data=df, x=col, bins=20, ax=axes[i])
    axes[i].set_title(f'{col} Distribution')
    axes[i].set_ylabel('Frequency')

# Remove empty subplots if any
for j in range(len(columns), len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

```



Analyzing data using groupBy

data.dtypes

id	object
name	object
host_id	object
host_name	object
neighbourhood_group	object
neighbourhood	object
latitude	float64
longitude	float64
room_type	object
price	float64
minimum_nights	float64
number_of_reviews	float64
last_review	object
reviews_per_month	float64
calculated_host_listings_count	float64
availability_365	float64
number_of_reviews_ltm	float64

```

license          object
rating           object
bedrooms         object
beds             int64
baths            object
dtype: object

```

```
df.groupby('neighbourhood_group')['price'].mean()
```

```

neighbourhood_group
Bronx          107.990506
Brooklyn       155.138317
Manhattan      204.146014
Queens         121.681939
Staten Island  118.780069
Name: price, dtype: float64

```

```
# Avg price per bed
```

```
df.groupby('neighbourhood_group')['price_per_bed'].mean()
```

```

neighbourhood_group
Bronx          74.713639
Brooklyn       99.788493
Manhattan      138.708057
Queens         76.336210
Staten Island  67.728101
Name: price_per_bed, dtype: float64

```

Feature Engineering

```

#Per bed price
# ['price_per_bed']
df['price_per_bed']=df['price']/df['beds']

```

```

C:\Users\amkoo\AppData\Local\Temp\ipykernel_30476\1716971920.py:3:
SettingWithCopyWarning:

```

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df.head()
```

```

          id
name \
0      1312228.0      Rental unit in Brooklyn · ★5.0 · 1

```

```

bedroom
1          45277537.0  Rental unit in New York · ★4.67 · 2 bedrooms
....
2  971000000000000000.0  Rental unit in New York · ★4.17 · 1 bedroom ·
...
3          3857863.0  Rental unit in New York · ★4.64 · 1 bedroom ·
...
4          40896611.0  Condo in New York · ★4.91 · Studio · 1 bed ·
1...

```

```

      host_id      host_name neighbourhood_group
neighbourhood \
0    7130382      Walter      Brooklyn      Clinton
Hill
1    51501835      Jeniffer      Manhattan      Hell's
Kitchen
2    528871354      Joshua      Manhattan
Chelsea
3    19902271  John And Catherine      Manhattan      Washington
Heights
4    61391963      Stay With Vibe      Manhattan      Murray
Hill

```

```

      latitude longitude      room_type  price  ...
reviews_per_month \
0  40.683710 -73.964610      Private room  55.0  ...
0.03
1  40.766610 -73.988100  Entire home/apt  144.0  ...
0.24
2  40.750764 -73.994605  Entire home/apt  187.0  ...
1.67
3  40.835600 -73.942500      Private room  120.0  ...
1.38
4  40.751120 -73.978600  Entire home/apt  85.0  ...
0.24

```

```

      calculated_host_listings_count availability_365
number_of_reviews_ltm \
0          1.0          0.0
0.0
1          139.0          364.0
2.0
2          1.0          343.0
6.0
3          2.0          363.0
12.0
4          133.0          335.0
3.0

```

```

      license  rating  bedrooms  beds      baths  price_per_bed

```

0	No License	5	1	1	Not specified	55.0
1	No License	4.67	2	1	1	144.0
2	Exempt	4.17	1	2	1	93.5
3	No License	4.64	1	1	1	120.0
4	No License	4.91	Studio	1	1	85.0

[5 rows x 23 columns]

Bivariate Analysis: One Variable depends on another variable

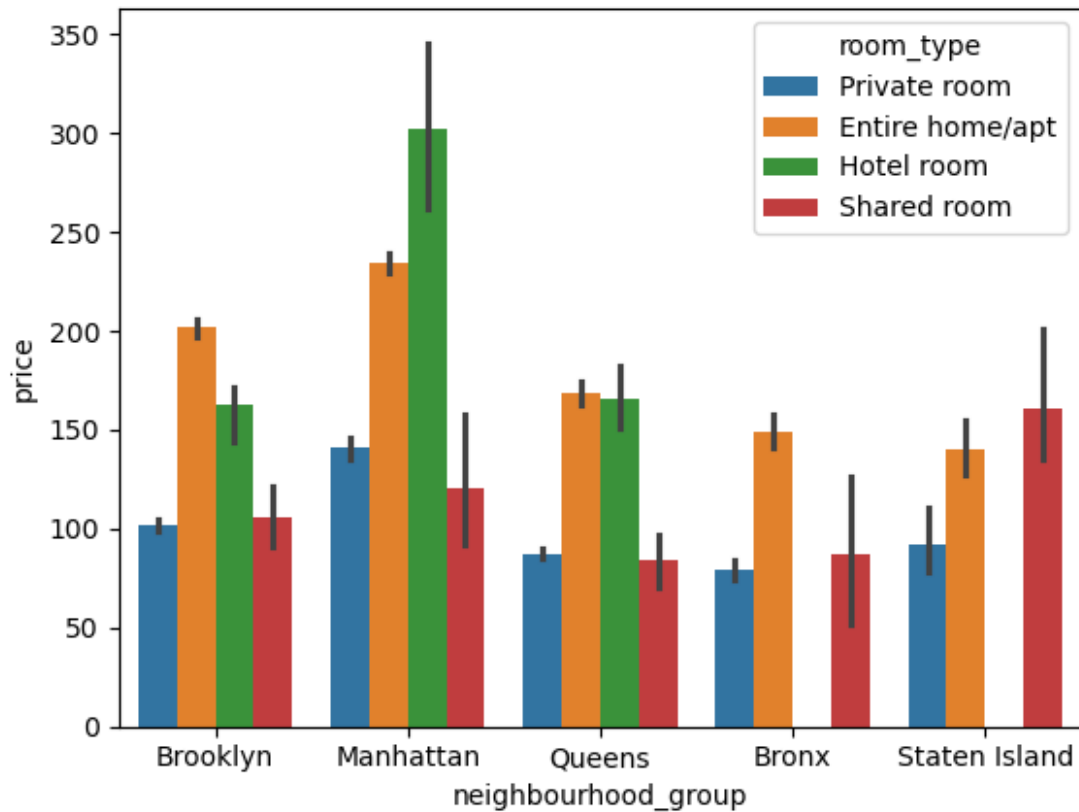
```
df.columns
```

```
Index(['id', 'name', 'host_id', 'host_name', 'neighbourhood_group',
       'neighbourhood', 'latitude', 'longitude', 'room_type', 'price',
       'minimum_nights', 'number_of_reviews', 'last_review',
       'reviews_per_month', 'calculated_host_listings_count',
       'availability_365', 'number_of_reviews_ltm', 'license',
       'rating',
       'bedrooms', 'beds', 'baths', 'price_per_bed'],
      dtype='object')
```

```
# Price depended on neighbourhood
```

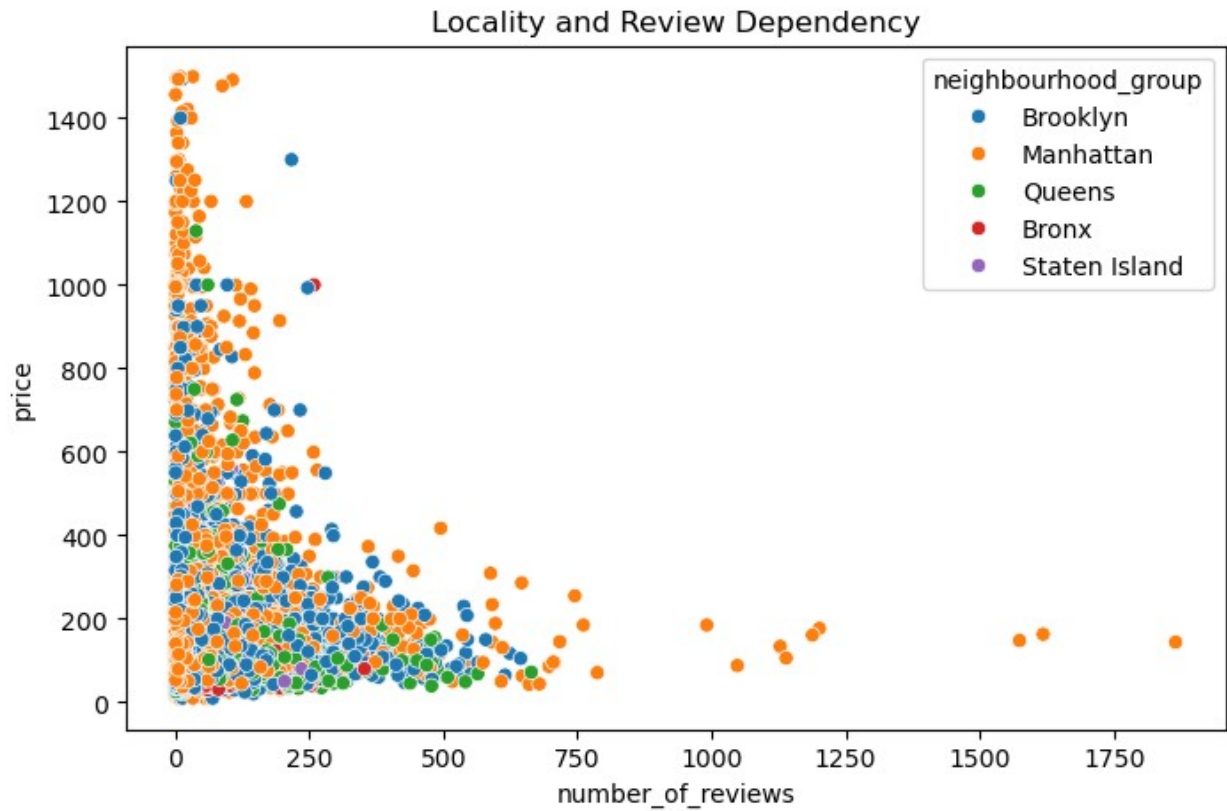
```
sns.barplot(data=df, x='neighbourhood_group', y='price', hue='room_type')
```

```
<Axes: xlabel='neighbourhood_group', ylabel='price'>
```

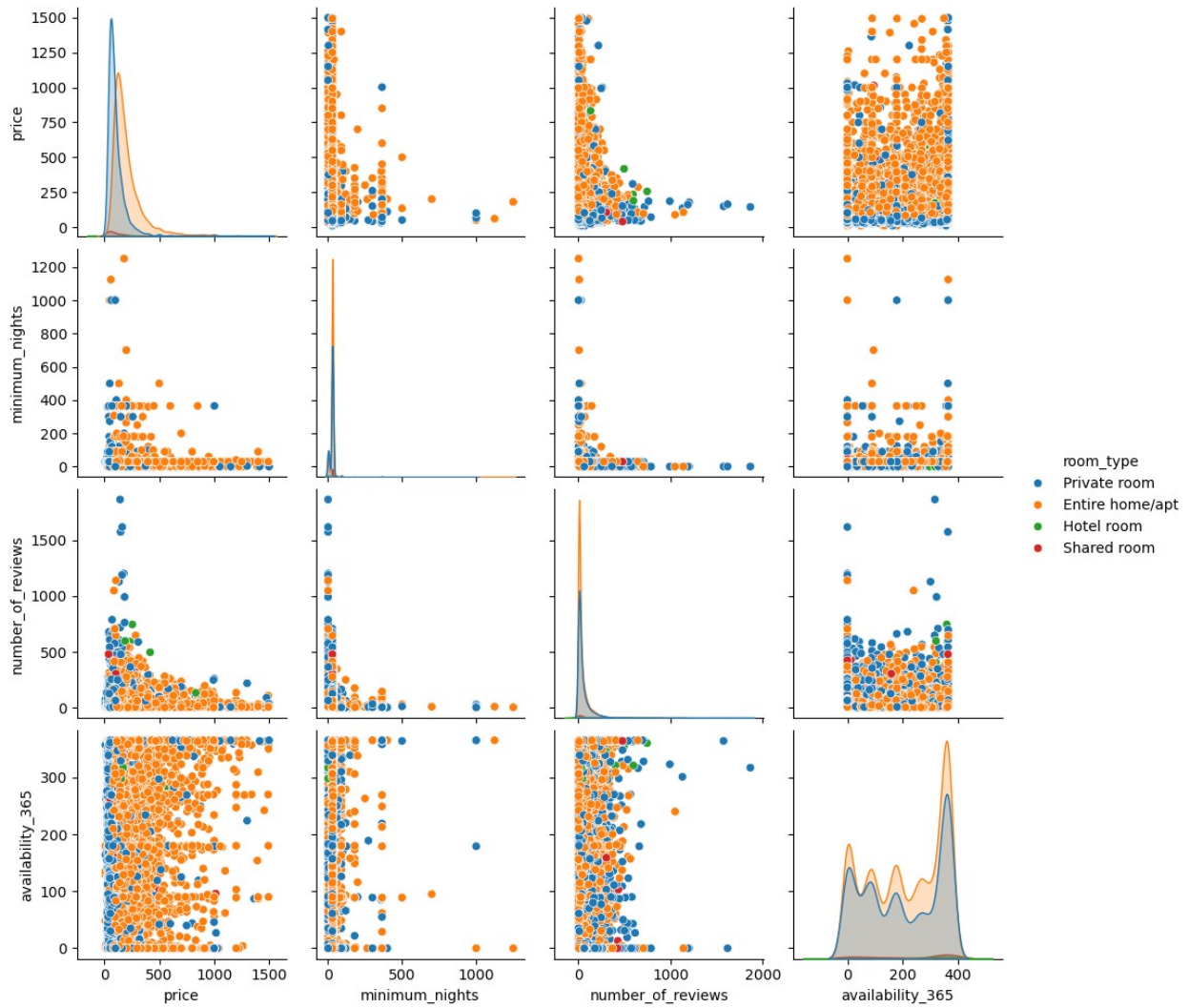



```
# Number of review with a price
plt.figure(figsize=(8,5))
plt.title('Locality and Review Dependency')
sns.scatterplot(data=df,x='number_of_reviews',y='price',hue='neighbourhood_group')

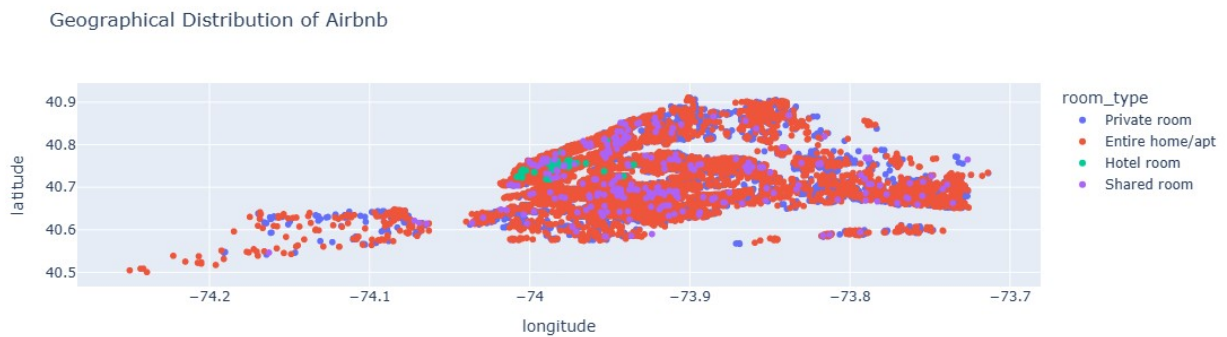
<Axes: title={'center': 'Locality and Review Dependency'},
xlabel='number_of_reviews', ylabel='price'>
```



```
sns.pairplot(data=df, vars=['price', 'minimum_nights', 'number_of_reviews', 'availability_365'], hue='room_type')  
<seaborn.axisgrid.PairGrid at 0x13d3d0bc2f0>
```



```
# Geographical Distribution of Airbnb
colortype=df['room_type']
px.scatter(df,x='longitude',y='latitude',color=colortype,title="Geographical Distribution of Airbnb")
```



Understanding correlation i.e one column to another that is only for number datatypes

Heatmap:Correlation of one variable with others for numerical column data.dtypes

id	object
name	object
host_id	object
host_name	object
neighbourhood_group	object
neighbourhood	object
latitude	float64
longitude	float64
room_type	object
price	float64
minimum_nights	float64
number_of_reviews	float64
last_review	object
reviews_per_month	float64
calculated_host_listings_count	float64
availability_365	float64
number_of_reviews_ltm	float64
license	object
rating	object
bedrooms	object
beds	int64
baths	object
dtype:	object

```
corr=df[['latitude','longitude','price','minimum_nights','number_of_reviews','reviews_per_month','availability_365','beds']].corr()  
corr
```

	latitude	longitude	price	minimum_nights	\
latitude	1.000000	0.047369	0.012686	0.004590	
longitude	0.047369	1.000000	-0.193728	0.023890	
price	0.012686	-0.193728	1.000000	-0.044635	
minimum_nights	0.004590	0.023890	-0.044635	1.000000	
number_of_reviews	-0.047953	0.004820	-0.043533	-0.059049	
reviews_per_month	-0.041673	0.041720	-0.012775	-0.122509	
availability_365	-0.005941	0.063523	0.048036	0.035466	
beds	-0.071753	0.041832	0.415278	-0.025852	

	number_of_reviews	reviews_per_month	
availability_365 \			
latitude	-0.047953	-0.041673	-
0.005941			
longitude	0.004820	0.041720	
0.063523			

price	-0.043533	-0.012775	
0.048036			
minimum_nights	-0.059049	-0.122509	
0.035466			
number_of_reviews	1.000000	0.631005	-
0.049656			
reviews_per_month	0.631005	1.000000	-
0.040116			
availability_365	-0.049656	-0.040116	
1.000000			
beds	0.040071	0.053496	
0.065985			

	beds
latitude	-0.071753
longitude	0.041832
price	0.415278
minimum_nights	-0.025852
number_of_reviews	0.040071
reviews_per_month	0.053496
availability_365	0.065985
beds	1.000000

```
plt.figure(figsize=(8,6))
sns.heatmap(data=corr,annot=True)
```

<Axes: >

